



UNIVERSIDAD
NACIONAL
DE COLOMBIA

Prototype 4 Quality Attributes, Part 2

Jeisson Andrés Vergara Vargas

Software Architecture

2026-I

1. Objective

The objective of the fourth delivery of the project is to **redesign the architecture** and generate a final **prototype** of the software system developed as a project.

2. Requirements

2.1. Functional Requirements

- The functional completeness for the software system must be defined by the team.

2.2. Non-Functional Requirements

- The software system must respond to two different **performance** and **scalability** scenarios:
 - In scenario 1: the software system must implement the **Load Balancer Pattern**. Performance testing is required.
 - In scenario 2: the software system must implement a pattern defined by the team.
- The software system must respond to at least four different **reliability (high availability, resilience or fault tolerance)** scenarios.
 - In scenario 1: the software system must implement the **Replication Pattern ({Hot, Warm, Cold} Spare)**.
 - In scenario 2: the software system must implement the **Service Discovery Pattern**.
 - In scenario 3: the software system must implement the **Cluster Pattern**.
 - In scenario 4: the software system must implement a pattern defined by the team.
- The software system must respond to an **interoperability** scenario.
- **Security scenarios** of Prototype 3 must be ensured again during the architectural redesign to be carried out for reliability.

3. Delivery

3.1. Artifact

- Team
 - Name (1a, 1b, ..., 2a, 2b, ...)
 - Full names of the team members.
- Software System
 - Name
 - Logo
 - Description
- Architectural Structures
 - Component-and Connector (C&C) Structure
 - C&C View
 - Description of architectural elements and relations.
 - Description of architectural styles and patterns used.
 - Deployment Structure
 - Deployment View
 - Description of architectural elements and relations.
 - Description of architectural patterns used.
 - Layered Structure
 - Layered View
 - Description of architectural elements and relations.
 - Description of architectural patterns used.
 - Decomposition Structure
 - Decomposition View
 - Description of architectural elements and relations.
- Quality Attributes
 - Security
 - Security scenarios.
 - Applied architectural tactics.
 - Applied architectural patterns.
 - Performance and Scalability
 - Performance scenarios.
 - Applied architectural tactics.
 - Applied architectural patterns.
 - Performance testing analysis and results.
 - Reliability
 - Reliability (high availability, resilience or fault tolerance) scenarios.
 - Applied architectural tactics.
 - Applied architectural patterns.
 - Interoperability
 - Interoperability scenario.
 - Applied architectural tactics.
 - Applied architectural patterns.

3.2. Submission Instructions

- The artifact must be created in **.md** format and exported to **.pdf** format.
- The artifact must be named **p4_X.pdf** (where X = the team name [1a, 1b, ..., 2a, 2b, ...]).
- The artifact must be submitted via [MiCampus UNAL](#).
- Only one member of the team must submit the artifact.

3.3. Delivery Deadline

Wednesday, June 10, 2026.

3.4. Delivery Presentation

Thursday, June 11, 2026 (in class).